



A data-driven machine learning approach to predict desalination capacities of faradic materials for capacitive deionization

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ABSTRACT

Capacitive deionization (CDI) stands out as a promising desalination method owing to its energy efficiency, environmental sustainability, and high performance. Faradic materials, with high theoretical capacities and abundant redox reaction sites, have shown substantial potential for practical CDI applications. However, the exact impact of material properties and operational conditions on desalination capacity of faradic materials remains unclear, requiring numerous time-consuming and inefficient CDI experiments. Machine learning (ML) offers a prospective solution by training large amounts of data to provide a more efficient approach for predicting the desalination capacity of faradic materials with minimal investment in time and resource. This work employed four ML models to predict the desalination capacity of faradic materials and investigate the relationships between various features and desalination capacity. Notably, the gradient boosting decision tree model achieved high prediction accuracy with a mean absolute error of 6.34 mg g^{-1} and root mean square error of 8.58 mg g^{-1} . SHapley Additive exPlanations analysis identified specific capacitance as the most influential feature for desalination capacity. Finally, Prussian blue analogues and molybdenum disulfide, as representative faradic materials, were prepared and tested for CDI, confirming the effectiveness of ML predictions. This study demonstrates the excellent function of ML in predicting the desalination capacity of faradic materials and highlights its potential to accelerate the development of advanced CDI systems.

1. Introduction

With a growing population and increasing environmental pollution, the acquisition of freshwater resources has become a critical issue. Currently, various desalination technologies serve as crucial means of obtaining freshwater. However, traditional techniques, such as electro-dialysis and reverse osmosis, are associated with high energy consumption and significant operational costs [1,2]. To solve these limitations, capacitive deionization (CDI) has emerged as an effective technology for addressing freshwater shortages owing to its merits of energy efficiency, environmental friendliness, and high performance [3–8]. Notably, the choice of electrode materials plays a crucial role in

CDI technology, directly affecting its desalination capacity [9–11]. Traditional carbon materials such as activated carbon, carbon nanofibers, and carbon nanotubes have primarily been used in the CDI field [12–16]. However, the limited desalination capacity of these carbon materials has severely restricted the further development of carbon-based CDI systems [17–19]. To further improve the desalination capacity of CDI, faradic materials, which possess high theoretical capacities and abundant redox reaction sites, have been introduced to replace traditional carbon materials [20–25]. Although existing faradic materials have demonstrated outstanding desalination capacity, there is still a need for novel faradic materials with even higher capacity to meet practical application demands. However, traditional methods for

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exploring faradic materials, which are time-consuming and labor-intensive, cannot effectively regulate desalination capacity in a controllable manner and often require extensive trial and error.

Recently, machine learning (ML) has emerged as a significant technology in the fields of batteries and supercapacitors, owing to its ability to predict the properties and performance of electrode materials [26–29]. This compensates for the limitations of traditional experimental methods. Furthermore, ML provides an effective means of clarifying the obscure linkages between material properties and performance [30–32]. Its operational mechanism primarily involves training various ML models with large datasets to achieve accurate predictions, thereby making ML applicable to the CDI field as well [33–39]. Recent studies in the CDI field demonstrate that ML can not only forecast the adsorption performance of carbon materials but also reveal the underlying correlations between their properties and performance. For example, Zhang et al. employed random forest (RF) and artificial neural network (ANN) models to forecast the desalination capacity and rate of carbon materials, further elucidating the ambiguous linkages between material features and desalination performance [40]. Cho et al. utilized an RF model to forecast the desalination capacity of carbon materials and explored the importance of operational conditions and electrode properties [41]. In our previous work, we employed four different ML algorithms to forecast the salt adsorption capacity (SAC) and rate of porous carbon with high accuracy, and importantly, these ML predictions were further validated through specific CDI experiments [42]. Notably, these studies primarily focus on the desalination performance of carbon materials. There still remains a lack of related investigations via ML approaches in predicting SAC for faradic materials, as well as the interdependencies among their properties, operational conditions, and SAC, which is highly helpful for exploring high-performance faradic materials.

Herein, we applied ML to predict the desalination capacity of faradic materials and further explore the relationship between various features and their desalination capacity. Specifically, four key features were considered as inputs: specific capacitance (Cs), specific surface area (SSA), applied voltage (AV), and electrolyte concentration (EC). To accurately predict the SAC of faradic materials, four ML models were employed: gradient boosting decision tree (GBDT), support vector machine (SVM), ANN, and RF. Fig. 1 shows the prediction process of various ML models for the desalination capacity of faradic materials. Notably, the GBDT model delivered superior accuracy with mean absolute error (MAE) value of 6.34 mg g^{-1} and root mean square error (RMSE) value of 8.58 mg g^{-1} . Subsequently, the relative importance of

these four features to SAC was evaluated using SHapley Additive exPlanations (SHAP) analysis. Cs exerted the most influential role in determining desalination capacity and exhibited positive correlation with desalination capacity. Finally, Prussian blue analogues (PBAs) and molybdenum disulfide (MoS_2), representing the two most important types of faradic materials—redox-active materials and pseudocapacitive materials, respectively—were prepared for CDI tests to further validate the ML predictions because of their abundant active sites and high theoretical capacity [43–46]. The experimental results demonstrated consistency between the ML predictions and the CDI test outcomes, confirming the successful extension of ML-guided predictions of desalination capacity to the field of faradic materials. This study not only showcases the versatile nature of ML methods but also fills a critical gap in their application to faradic materials.

2. Methods

2.1. Data collection and feature selection

The dataset utilized in this work contains 510 samples obtained from previous works, encompassing various faradic materials, including redox-active materials and pseudocapacitive materials. Detailed information about the dataset is provided in Table S1 (Supporting Information). In this dataset, SAC is utilized as the prediction target. The classic material properties, Cs and SSA, regarded to effectively influence the desalination capacity of electrode materials were used as input features for the ML models. Additionally, operational conditions such as EC and AV affect the contact between electrodes and salt ions, as well as the driving force used in the adsorption process, respectively. To identify the optimal ML model for accurately predicting the desalination capacity and further analyze feature importance, four commonly used features (Cs, SSA, EC, and AV) were employed as input features to train four different ML models (GBDT, RF, ANN, and SVM).

2.2. ML models

In this work, four ML models: RF, GBDT, SVM, and ANN were selected to predict the SAC of faradic materials, mainly based on the following considerations: 1) Representative model types. The selected models represent three major categories of ML approaches: ensemble tree-based methods (RF and GBDT), kernel-based method (SVM), and neural network-based method (ANN). These models are widely used in materials science and offer complementary strengths in terms of

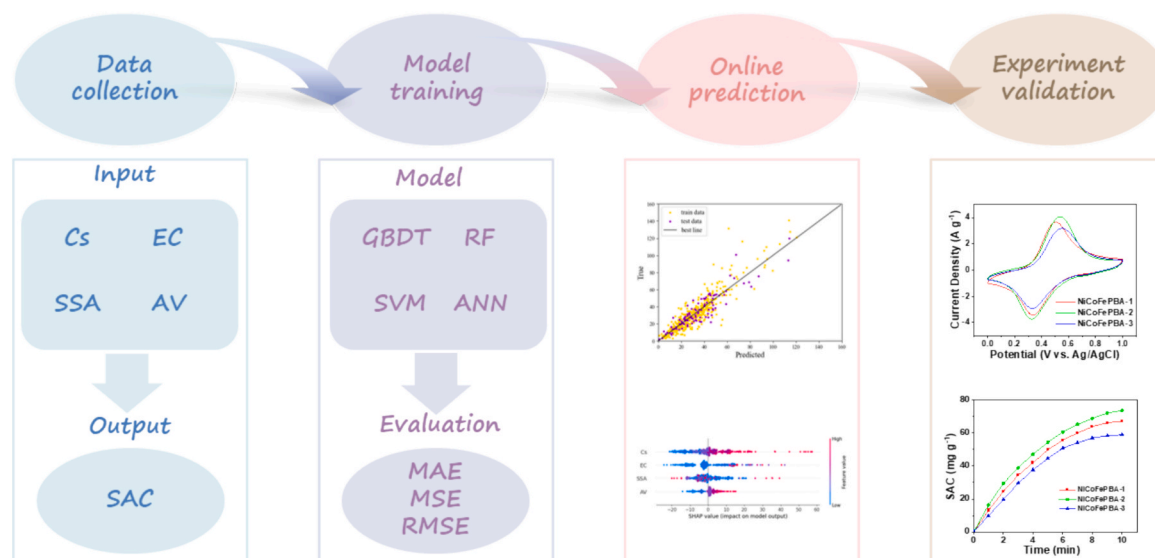


Fig. 1. The prediction process of ML for the desalination capacity of faradic materials.

interpretability, generalization, and nonlinear modeling, making them representative of different modeling paradigms. 2) Alignment with the requirements of materials performance prediction. In the context of predicting the SAC of faradic materials, multiple variables are often involved, with potentially complex and nonlinear interdependencies. The selected models offer diverse modeling mechanisms, which help ensure that the predictive framework is well-matched to the complexity of the task. 3) Proven effectiveness in prior materials research. These four models have been widely adopted in recent materials science literature [47–49]. In summary, the selection of these four models is based on their theoretical diversity, complementary strengths, and proven relevance to materials science problems. This ensures both the representativeness of our modeling approach and the scientific validity of the final model selection.

To further identify the optimal model for predicting the SAC of faradic materials, these four common ML models are compared in this work. As a classical algorithm, SVM is effective for solving both regression and classification problems. The core idea of SVM is to discover a hyperplane that maximally separates the samples, which is formulated as a convex quadratic programming problem [50]. ANNs, a prominent type of ML model, are extensively applied to tasks involving regression and classification. Structurally, they mimic neural behavior in the brain, consisting of input, hidden, and output layers. This structure allows ANN to deliver excellent prediction performance [51]. As an improved version of Bagging, RF applies random feature selection when constructing decision trees, where a subset of features is randomly sampled and the optimal feature is chosen for each node split. The averaging property of RF reduces variance, compensating for increased bias and thus resulting in a better overall model [52]. GBDT is an enhanced Boosting method. The core idea is to sequentially generate multiple weak learners, with each learner trained to fit the negative gradient of the loss function from the preceding model. This approach minimizes the model's total loss. GBDT incrementally and additively trains numerous models, effectively integrating many weak learners, which leads to excellent prediction accuracy [53]. Overall, different ML models have their own characteristics, and thus we need to select the most suitable ML model based on the dataset for predicting SAC.

2.3. Model training and evaluation

Four different ML models (GBDT, RF, ANN, and SVM) were employed in this work to predict SAC. A random partitioning of the dataset was performed, allocating 80 % for training and 20 % for testing. Each model was trained using five-fold cross-validation, and the optimized hyperparameters obtained were exhibited in Table S2. With these hyperparameters, the prediction performance of the models on the training and test sets was evaluated by the MAE, mean square error (MSE), and RMSE values. The specific equations are as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (1)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (2)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (3)$$

Where n indicates the number of samples, y_i and $\hat{y}_i$ represent the true and predicted values, respectively. Notably, smaller values of MAE, MSE, and RMSE indicate superior prediction performance of the model.

2.4. Experimental section

Ternary-metal PBAs (NiCoFe-PBAs), a representative class of redox-

active faradaic materials, were synthesized using a co-precipitation strategy. The specific synthesis process was as follows: different molar ratios of $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (1:5, 2:4, 3:3) were dissolved in 100 mL of 4.5 mmol $\text{C}_6\text{H}_5\text{Na}_3\text{O}_7 \cdot 2\text{H}_2\text{O}$ to form solution A, respectively. Solution B was prepared by dissolving 2 mmol of $\text{K}_3[\text{Fe}(\text{CN})_6]$ in 100 mL of H_2O , and then it was gradually added into solution A. After stirring for 5 min, the resulting solution was aged for 24 h. The precipitate was collected by centrifugation and subsequently dried. The final samples obtained were named as NiCoFePBA-1, NiCoFePBA-2 and NiCoFePBA-3, respectively.

MoS_2 , a representative class of pseudocapacitive faradaic materials, was provided from Sinopharm Chemical Reagent Co., Ltd. (Beijing, China). Defective MoS_2 (D- MoS_2) was obtained by thermally treating MoS_2 powders at 250 °C for 150 min.

The materials, characterization techniques and details of the CDI tests involved in this work are provided in the [Supporting Information](#).

3. Results and discussion

3.1. Spearman rank correlation coefficient study

In this study, the Spearman rank correlation coefficient study has been integrated into ML pipeline to analyze the correlation between features and targets [54]. Fig. 2 illustrates the Spearman rank correlation coefficient matrix for four features (EC, AV, SSA, Cs) and target (SAC). Each cell in the matrix shows the correlation coefficient between the respective variables, with the color intensity indicating the strength of the correlation. The Spearman rank correlation coefficient ranges from -1 to 1 , with positive and negative values representing positive and negative correlations, respectively. The results reveal a moderate positive correlation between EC and SAC (0.54) and between Cs and SAC (0.53). The positive correlation between EC and SAC can be attributed to the fact that higher EC provides more ions for the adsorption of faradic materials. Cs is a metric of the ability of an electrode material to store ions. Therefore, an electrode material with high specific capacitance can adsorb more ions, demonstrating a positive correlation between Cs and SAC. These findings are valuable for understanding the monotonic relationships between the variables, which is important for multivariable analysis.

3.2. Prediction of SAC

SAC is regarded as a significant metric for evaluating the desalination performance of faradic materials. Therefore, SAC as a target is accurately predicted by ML models in this work. Tables S3 and S4 show the prediction performances of four ML models on the training and test sets, respectively. Lower values of MAE, MSE, and RMSE indicate better performance of the ML models. Clearly, the GBDT model exhibits the best prediction accuracy for SAC, delivering the lowest MAE, MSE, and RMSE among the four ML models. Furthermore, the scatter plot (Fig. 3) displays the discrepancy between the predicted and true SAC for different ML models. The gold plots represent the training data, while the dark violet plots indicate the test data. Obviously, the gold and dark violet plots closely surround the best-fit line in the GBDT model, indicating the high prediction accuracy of the GBDT model. The superior performance of the GBDT model can be attributed to several key characteristics of the algorithm: 1) Sequential optimization mechanism. GBDT is an additive model that builds trees sequentially. Each new tree is trained to correct the prediction errors made by the previous ensemble. This gradient boosting process allows the model to progressively improve its accuracy by focusing on the residuals, leading to highly optimized predictions. 2) Strong nonlinear fitting capability. Compared to linear models or single decision trees, GBDT is particularly effective on capturing complex nonlinear relationships and feature interactions. This makes it well-suited for tasks where the underlying data patterns are intricate and not easily modeled by simpler algorithms. 3)

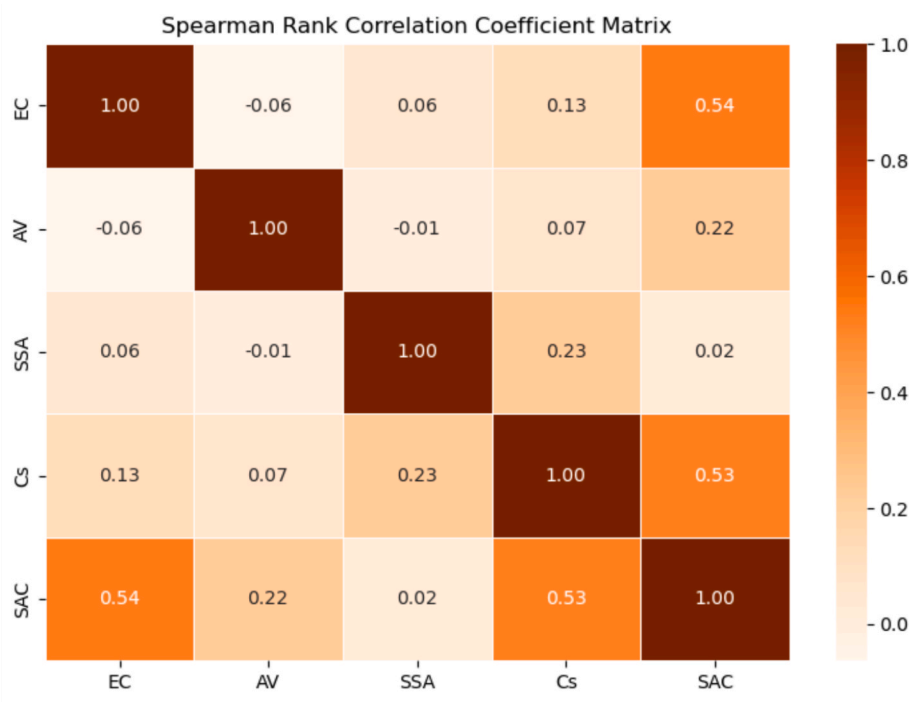


Fig. 2. Spearman rank correlation coefficient matrix including the features (EC, AV, SSA, Cs) and the target (SAC).

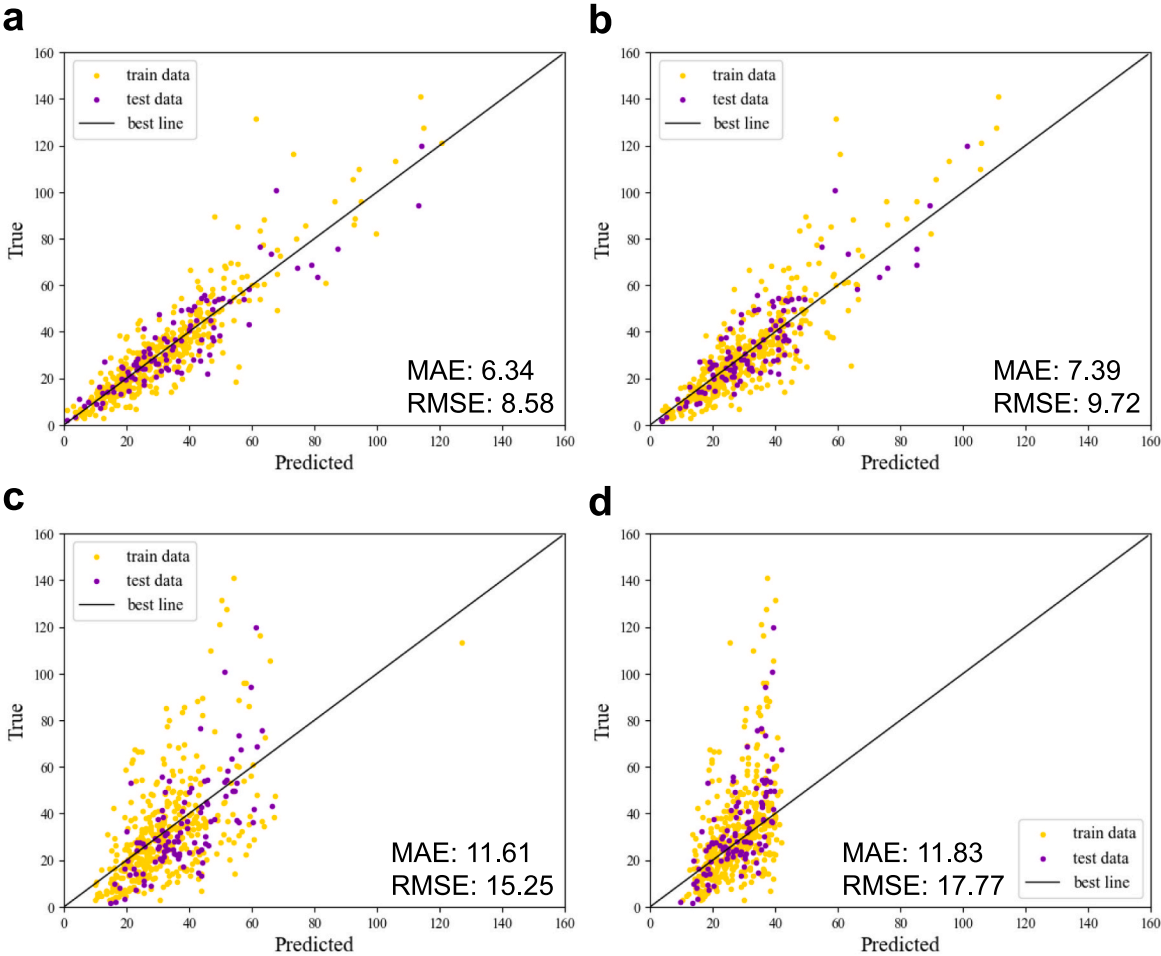


Fig. 3. Scatter plots of predicted SAC for (a) GBDT, (b) RF, (c) ANN and (d) SVM models.

Robustness to irrelevant features and outliers. GBDT inherently performs feature selection during training, reducing the influence of irrelevant variables. It is also relatively robust to outliers, which contributes to better generalization performance. 4) Good alignment with the characteristics of the dataset. The dataset used in this work is structured and tabular, with multiple features and potential nonlinear relationships. GBDT excels in handling such data, often outperforming models like ANN (which typically requires large-scale or unstructured data) or SVM (which can be sensitive to parameter tuning and scale poorly with large datasets). In summary, the superior performance of GBDT in this work is mainly due to its iterative error-correction strategy, strong ability to model complex patterns, and natural compatibility with structured data.

Additionally, the underlying mechanisms and rationale behind the selection of the best-performing ML model (GBDT) for predicting the desalination capacity of faradic materials in CDI systems are provided, as detailed below: 1) Compatibility between model characteristics and data features. The desalination performance of faradic materials is influenced by multiple factors, such as Cs, SSA, EC, and AV. These variables may exhibit complex nonlinear relationships and interactions. GBDT is particularly suitable for modeling such structured data with high-dimensional and interdependent variables due to its strong nonlinear fitting capability and built-in feature selection. 2) Incremental learning mechanism that captures subtle differences. In CDI systems, performance variations among faradic materials can be subtle—for instance, small improvements in capacity or stability. GBDT's sequential error-correcting learning process enables it to capture these fine-grained patterns more effectively than other models. 3) Model interpretability supports mechanism understanding. Despite being an ensemble model, GBDT allows for interpretability through SHAP analysis. This helps reveal which material properties most strongly influence performance, thereby supporting scientific reasoning and aiding in the rational design of materials—an important advantage for materials science research. 4) Empirical validation of model performance. As shown in Fig. 3, GBDT consistently outperforms other models (RF, SVM, ANN) across multiple evaluation metrics. Its superior predictive accuracy, consistency, and generalization ability provide strong empirical support for its selection as the most appropriate model for our task. In summary, the choice of GBDT is not arbitrary but is grounded in a solid understanding of both the algorithm's strengths and the nature of the data. This alignment between model capability and scientific problem enhances the reliability of the predictions and contributes meaningfully to the advancement of CDI material research.

3.3. Feature-importance analysis

ML models are often considered black boxes, and their non-interpretability limits the further development of ML in the CDI field [55]. Therefore, exploring novel interpretation methods to transform from black-box to transparent-box is of great significance for the

development of ML models. SHAP is introduced to overcome this limitation by providing a comprehensive visualization of how each feature contributes to the model's predictions [56,57]. The SHAP summary plot shown in Fig. 4 illustrates the impact of four features (Cs, EC, SSA, and AV) on the model's output (SAC). Each dot represents a data point, and the color gradient from blue to red helps in understanding whether low or high feature values contribute negatively or positively to the model's output. The x-axis shows the SHAP value, representing the feature's contribution to the prediction. The y-axis orders the features according to their importance, with those higher up deemed more significant. Notably, among the four features, Cs has the most significant influence on the prediction of SAC. Additionally, Cs shows a positive correlation with the prediction of SAC, which is consistent with the results of the Spearman rank correlation coefficient study. Notably, although this relationship had been observed in previous works, theoretical support had long been lacking. This finding, revealed through SHAP analysis, provides theoretical support for the positive correlation between Cs and SAC. Currently, ML has not been widely applied to faradic materials in CDI, primarily due to the limited availability of data in this field. With the theoretical foundation now established for the positive correlation between Cs and SAC, Cs can serve as a bridge to leverage the abundant data from the battery and supercapacitor fields. This approach can effectively expand the application of ML to faradic materials in the CDI field.

Conversely, SSA does not exhibit a strong positive or negative correlation with SAC but still has a notable effect on the prediction of SAC. The reasons for the different efficacies of Cs and SSA in predicting SAC are as follows: 1) For some faradic materials, their adsorption ability primarily relies on the intrinsic redox reactions of the materials themselves and is not significantly related to SSA [58]. Therefore, only Cs shows a positive correlation with the SAC of these faradic materials. As long as the faradic material has a high Cs, it can exhibit a high desalination capacity even with a low SSA. 2) For other faradic materials, an increase in SSA leads to some extent to an increase in Cs, thereby enhancing SAC [59]. In such a case, their SSA and Cs both show a positive correlation with their SAC. Conclusively, although different faradic materials exhibit different levels of feature importance for SAC prediction, Cs consistently shows a positive correlation with SAC prediction. EC displays a positive correlation for the prediction of SAC, consistent with the results of the Spearman rank correlation coefficient study. This is because higher EC provides more ions for the adsorption, improving the SAC of faradic materials. AV presents a positive correlation for predicting SAC, which is due to the fact that higher voltage can offer a stronger driving force for ion adsorption, thus benefiting the enhancement of SAC. These results from the ML methods align with theoretical expectations, paving the way for a new approach to study feature importance in the CDI field. In summary, Cs, as a representative material property, exhibits a more substantial impact than other features in predicting SAC. The importance of different features obtained from ML was further validated by the following CDI tests.

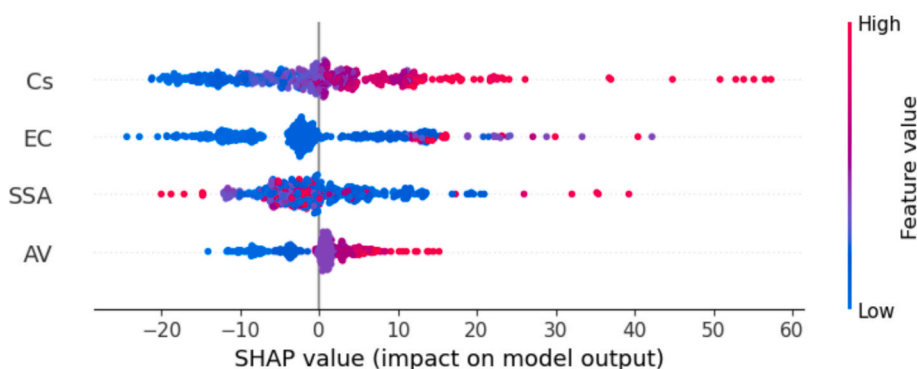


Fig. 4. Feature importance for predicted SAC using the SHAP method with the GBDT model.

3.4. Experiment validation

The X-ray diffraction (XRD) patterns of NiCoFe-PBA samples with different molar ratios of Ni: Co are shown in Fig. S1. Clearly, the diffraction peaks of three NiCoFe-PBA samples are similar and well match with the standard card (JCPDS card No. 82-2284), consisting with previous reported works and proving the successful synthesis of NiCoFe-PBA samples [60]. Fig. S2 presents the XRD patterns of MoS₂ and D-MoS₂, which display similar diffraction peaks. These peaks align well with the standard reference (JCPDS No. 37-1492) and are consistent with earlier published studies [61].

Furthermore, the nitrogen adsorption-desorption isotherms of NiCoFe-PBA samples are analyzed in Fig. 5a. All three NiCoFe-PBA samples exhibit typical type-IV sorption isotherms, indicating mesoporous-dominated structures. The SSA clearly rises with the increase of Ni²⁺ content, with values of 133.6, 159, and 200.2 m² g⁻¹ for NiCoFePBA-1, NiCoFePBA-2, and NiCoFePBA-3, respectively. This increase is attributed to the fact that higher Ni²⁺ content can reduce the grain size of NiCoFe-PBA samples [62]. As shown in Fig. 5b, both MoS₂ and D-MoS₂ exhibited typical type-IV sorption isotherms, with SSAs of 192.3 and 222.4 m² g⁻¹, respectively. The increase in SSA for D-MoS₂ can be attributed to the generation of abundant defects during the thermal treatment [63]. Moreover, the SSA and pore volume of all samples exhibit a similar trend, as shown in Fig. S3 and Table S5.

The electrochemical properties of different samples were tested by cyclic voltammetry (CV). Fig. 6a exhibits the CV curves of NiCoFe-PBA samples in 1 mol L⁻¹ NaCl solution at 5 mV s⁻¹. Distinct redox peaks appear in the CV curves, indicating the intercalation and extraction of Na⁺. Furthermore, the Cs of the three NiCoFe-PBA samples was calculated, as exhibited in Fig. S4. It is clear that with the enhancement of Ni content, the Cs of different samples first increases and then decreases. NiCoFePBA-2 delivers the highest Cs (253.3F g⁻¹) compared to NiCoFePBA-1 (235.1F g⁻¹) and NiCoFePBA-3 (208.6F g⁻¹), which is attributed to the optimal ratio of Ni: Co [64]. Fig. 6b presents the CV curves of MoS₂ and D-MoS₂ in 1 mol L⁻¹ NaCl solution at 5 mV s⁻¹. Clearly, all CV curves are nearly rectangular, indicating their pseudocapacitive behavior. As shown in Fig. S5, the Cs of D-MoS₂ (31.9F g⁻¹) is higher than that of MoS₂ (23.8F g⁻¹), which can be attributed to the formation of abundant defects on the surface of D-MoS₂.

The desalination performance of different samples was evaluated to validate the predictions made by ML models. As illustrated in Fig. 7a, the SAC over time for different NiCoFe-PBA samples was measured in a 1500 mg L⁻¹ NaCl solution at 1.2 V. Notably, the NiCoFePBA-2 electrode achieved the highest SAC of 73.8 mg g⁻¹, surpassing the SAC values of NiCoFePBA-1 (67.2 mg g⁻¹) and NiCoFePBA-3 (59 mg g⁻¹). This finding suggests that redox-active electrodes with higher Cs possess a greater capacity for ion adsorption, which aligns with the ML predictions. Furthermore, the results indicate no clear correlation between SSA and

SAC, which can be attributed to the fact that the adsorption mechanism of redox-active faradic materials primarily relies on the intrinsic redox reactions of the materials themselves. As shown in Fig. 7b, the D-MoS₂ electrode delivered a higher SAC of 40.2 mg g⁻¹ compared to that of MoS₂ (25.8 mg g⁻¹). It was observed that pseudocapacitive electrodes with higher Cs and SSA tend to exhibit higher SAC values, indicating a positive correlation between both SSA and Cs with SAC. Overall, for different types of faradic materials, although SSA shows varying positive or negative correlations with SAC, Cs consistently exhibits a positive correlation with SAC, which is consistent with the ML predictions. Additionally, as shown in Fig. 8a, NiCoFePBA-2 demonstrated a higher SAC in a 1500 mg L⁻¹ NaCl solution compared to lower concentrations of 500 mg L⁻¹ and 1000 mg L⁻¹, suggesting that increased EC facilitates greater ion availability for adsorption. Fig. 8b further illustrates the SAC variation of NiCoFePBA-2 at different voltages in a 1500 mg L⁻¹ NaCl solution, revealing a significant enhancement in SAC with increasing voltage. This indicates that higher AV provides additional energy for the adsorption process. Collectively, these findings underscore the distinct yet significant influences of Cs, SSA, EC, and AV on the SAC of faradic materials, corroborating the SHAP results and validating the prediction accuracy of the ML methodologies employed.

4. Conclusions

In this work, we applied four ML models (GBDT, RF, ANN, SVM) to accurately forecast the SAC of faradic materials and explore the relationships between key features and desalination capacity. Notably, the GBDT model demonstrated high accuracy with a low MAE value of 6.34 mg g⁻¹ and a low RMSE value of 8.58 mg g⁻¹. SHAP analysis identified Cs as the most influential factor in determining SAC, hinting that Cs is the most deserving factor in the study and design of novel faradic materials in the future. More importantly, this finding provides theoretical support for the positive correlation between Cs and SAC. Due to limited data, ML has not been widely applied to faradic materials in CDI. In future research, Cs can serve as a bridge to leverage abundant battery and supercapacitor data, expanding ML applications to faradic materials in CDI. Furthermore, ternary-metal PBAs and MoS₂, representing the two most important types of faradic materials—redox-active and pseudocapacitive materials, respectively—were systematically synthesized and tested to validate the ML prediction results, highlighting the significance of Cs for desalination capacity and confirming the reliability and applicability of ML. In the future, with the increasing application of faradic materials in the CDI field, the dataset will become even larger, benefiting enhanced prediction accuracy of ML models. Overall, we showcased the promising role of ML in accelerating both the development and performance tuning of faradaic materials for CDI systems, paving the way for more efficient and effective CDI desalination technology.

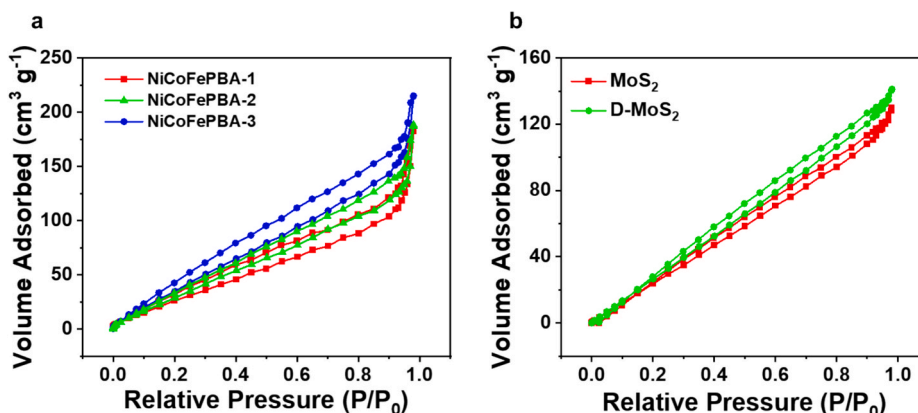


Fig. 5. Nitrogen sorption isotherms of (a) different NiCoFe-PBA samples and (b) MoS₂ and D-MoS₂.

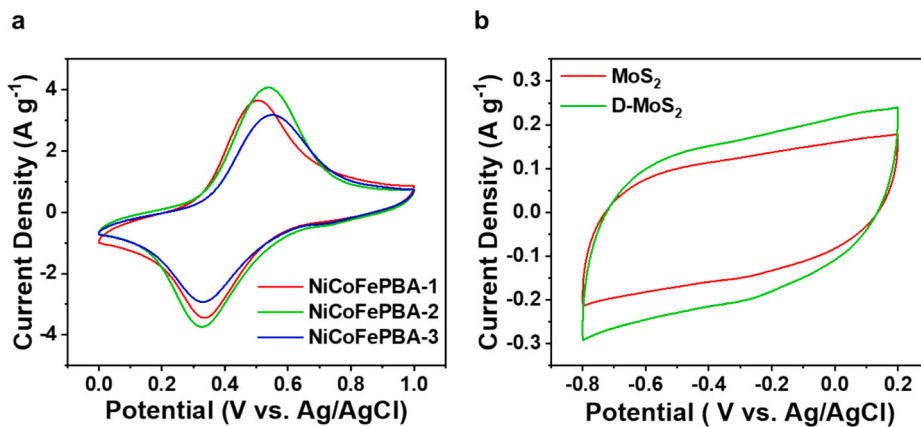


Fig. 6. CV curves of (a) different NiCoFe-PBA samples and (b) MoS₂ and D-MoS₂.

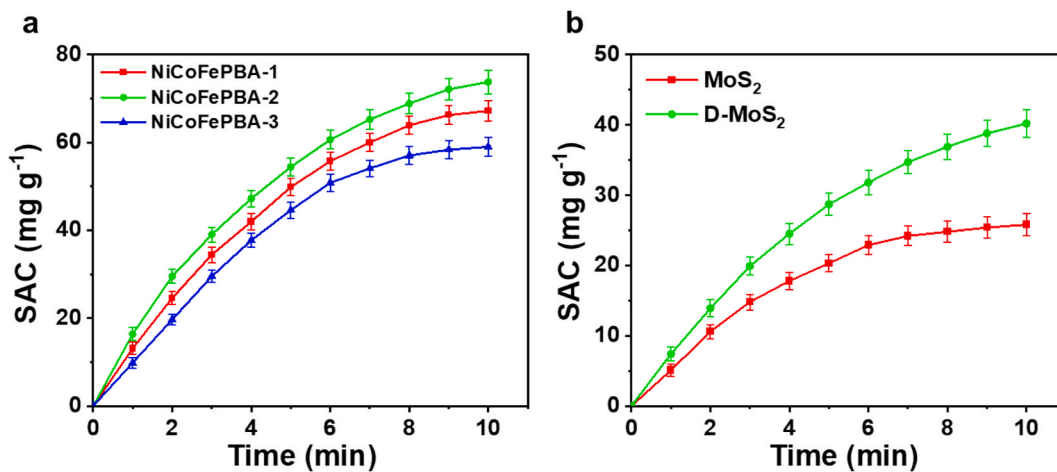


Fig. 7. The curves of (a) SAC with time of different NiCoFe-PBA samples and (b) SAC with time of MoS₂ and D-MoS₂ in a 1500 mg L⁻¹ NaCl solution at 1.2 V. Error bars represent the relative standard deviation (n = 3).

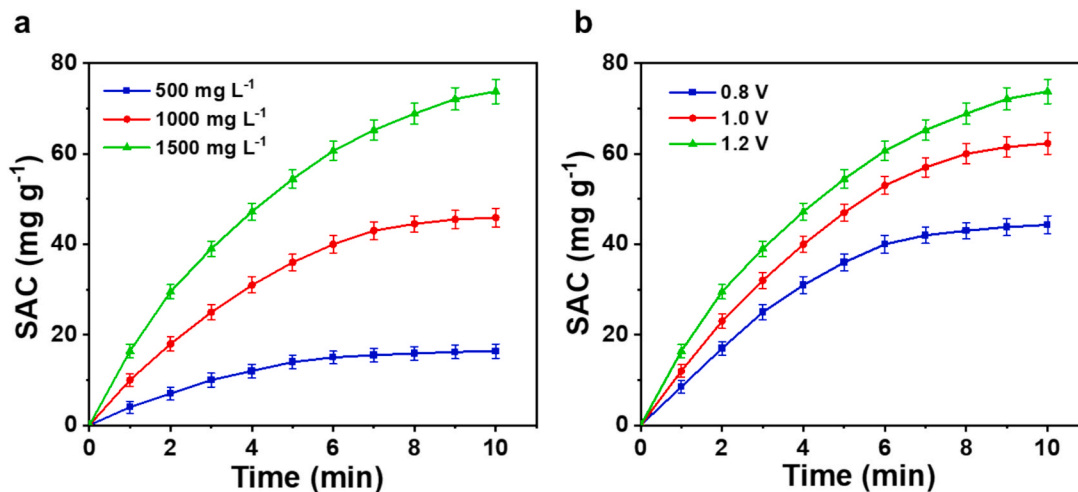


Fig. 8. SAC curves of (a) NiCoFePBA-2 at 1.2 V with different EC and (b) NiCoFePBA-2 in 1500 mg L⁻¹ NaCl solution at different AV. Error bars represent the relative standard deviation (n = 3).

CRediT authorship contribution statement

Hao Wang: Conceptualization, Methodology, Writing – original draft. **Chenglong Wang:** Validation, Writing – review & editing.

Junfeng Li: Visualization. **Yuquan Li:** Formal analysis. **Yong Liu:** Investigation, Writing – review & editing. **Zequi Chen:** Data curation. **Xinjuan Liu:** Formal analysis. **Guang Yang:** Investigation. **Likun Pan:** Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.seppur.2025.134181>.

Data availability

Data will be made available on request.

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